Stop Interpolation Topologically

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Agenda

Introduction

Persistent Homology

Triangulation of Manifolds

Persistence Stability

Natural Neighbor Interpolation

The Collapsing Sequence

Hypothesis Testing

Summary

Introduction

Persistent Homology

An outline to persistent homology

A brief description

Persistent homology is the study of k-**dimensional abelian groups** attached to a topological space. They are studied algebraically and behave *similar* to the topological space itself under change revealing properties of the original space they are attached to by their *invariants*.

The connection of abelian groups by a boundary homomorphism allows to investigate all k-dimensions at once. The induced object is a chain complex.

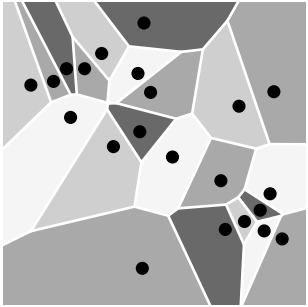
Triangulation of Manifolds

Persistence Stability

Natural Neighbor Interpolation

Step 1: Voronoi diagram

Natural neighbor interpolation example

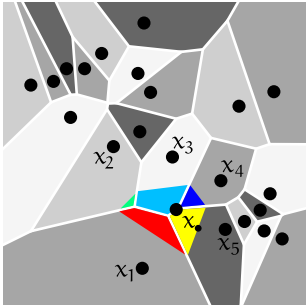


The **Voronoi Diagram** is spanned over the point cloud embedded into some Euclidean space. Technically one does not calculate the Voronoi diagram but the dual **Delaunay triangulation**.

$$\text{Vordgm} = \{x \in X \mid d(x, \lambda_i) \leq d(x, \lambda_j)\} \quad (1)$$

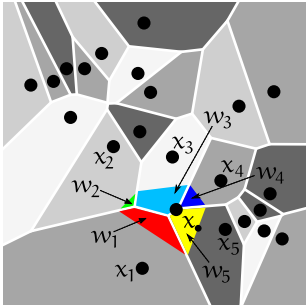
Step 2: Add a point into the convex hull

We need to compute the weights per region



An outline to persistent homology

A brief description



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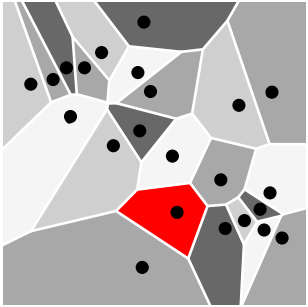

The Collapsing Sequence

Hypothesis Testing

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